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References

Migraine as a Differential Diagnosis of CADASIL in Young Individuals

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Background and Aims

Leukoencephalopathy, known worldwide as CADASIL. It is a disease caused by a mutation in the NOTCH3 gene on chromosome 19, which encodes a protein whose main function is to maintain vascular structural and functional stability. There are no universally accepted diagnostic criteria for CADASIL, as the presenting symptoms, age of onset and disease progression are variable. Although cerebrovascular lesions are the most frequent clinical manifestations, migraine is the earliest symptom, which may already be present in early adolescence.

Objective

To identify migraine as an initial symptom of CADASIL.

Methods

Assessment of migraine in patients with CADASIL through comprehensive literature review.

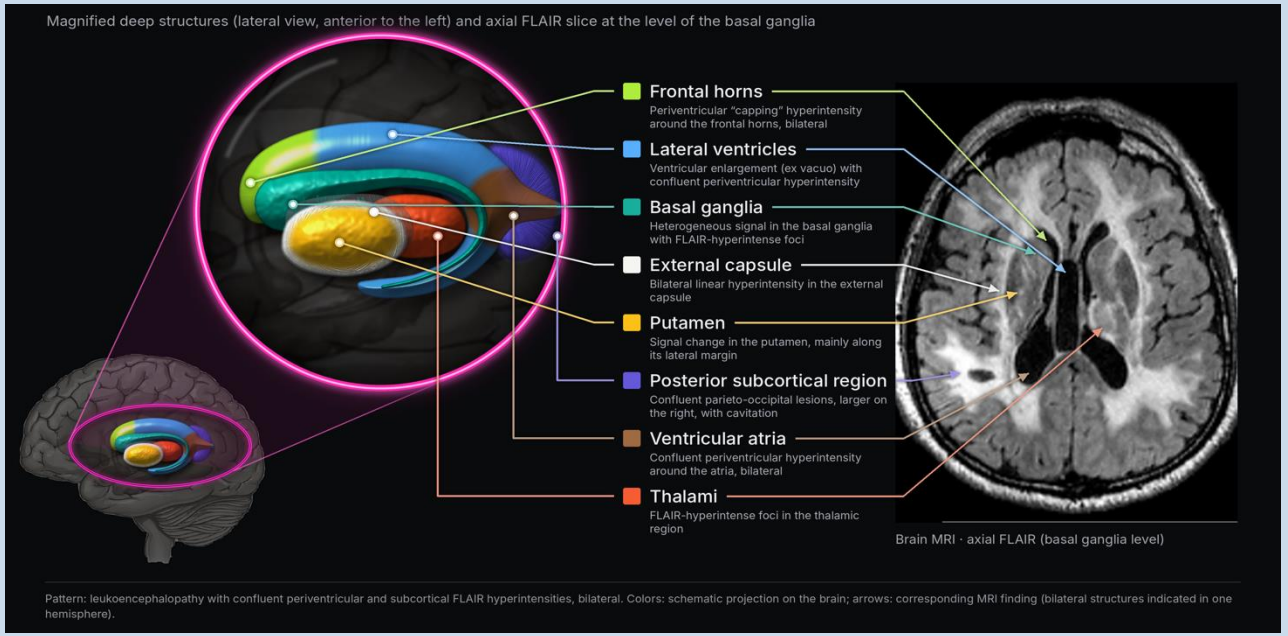


Figure 1: MRI findings in CADASIL showing characteristic white matter lesions and affected structures: anatomical × MRI (FLAIR) correlation

Clinical Relevance: Early recognition of migraine with aura in young patients, especially with family history of stroke or dementia, should prompt consideration of CADASIL as a differential diagnosis.

Results

The proportion of migraine in patients with CADASIL is five times higher than in the general population. The first symptoms usually occur before the age of 20, with the average age of onset being 30 years. In women, symptoms appear earlier than in men.

Conclusions

Migraine, especially when accompanied by aura symptoms, may be an initial symptom of CADASIL and should be considered as a differential diagnosis, particularly in younger individuals. Early identification of CADASIL through recognition of its association with migraine can lead to earlier diagnosis, better management of symptoms, and improved genetic counseling for affected families.